

Influence_Consensus_agr_dea - In society, does Agriculture alone generally have a direct effect on the death rate, or are observed relationships primarily indirect? If a direct effect exists, is it negative, none, negligible, or positive? What would be a reasonable normalized magnitude for this influence on a scale from -1.00 to 1.00? Please only provide evidence for only direct influences

Agriculture alone has a direct but complex influence on death rates, with evidence showing both positive and negative effects depending on the context and agricultural activity type. Studies in Northern Ireland indicate that higher farming intensity at the community level is associated with increased mortality risk, suggesting a direct positive effect on death rates in those areas (Trearty et al., 2024; Trearty et al., 2025; Trearty et al., 2023). Conversely, cohort studies of agricultural workers in the U.S. show lower overall mortality compared to the general population, although certain causes of death such as injuries related to machinery and specific cancers are elevated among farmers, indicating some direct occupational risks that increase death rates (Waggoner et al., 2010; Shrestha et al., 2019; Lower et al., 2022). In Nigeria, fishing (an agricultural component) directly reduces mortality, while forestry activities increase it; crop and livestock production showed no significant long-term direct effect on mortality (Olaniran & Ogunjinmi, 2022). Additionally, animal husbandry has been linked to higher suicide rates in Japan, pointing to a direct positive association between this type of agriculture and specific mortality causes (Kanamori & Kondo, 2019). Overall, the direct effect of agriculture on death rate is mixed but generally small to moderate in magnitude; a reasonable normalized estimate would range from about -0.1 (for protective effects like fishing) up to +0.3 (for increased risks from farming intensity or animal husbandry), reflecting context-dependent influences on mortality (Olaniran & Ogunjinmi, 2022; Trearty et al., 2024; Waggoner et al., 2010; Kanamori & Kondo, 2019).

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